

Performance Testing

Date	04 July 2026
Team ID	
Project Name	Rising water
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Testing & Basic Performance Testing
Target Module / API	Flood Prediction API (/predict) and Flask Web Application
Test Environment	Local (Windows 11, Flask Development Server)
Test Date	04-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid flood parameters	1	30	Prediction result displayed successfully
2	Submit invalid or empty input values	1	30	Validation error message displayed

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
3	Multiple prediction requests	20	60	System responds without errors
4	Continuous prediction requests	50	120	Stable response with acceptable performance

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 seconds	Pass	Meets performance target
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Acceptable under load
3	Throughput (Req/sec)	> 20 Req/sec	25 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	48%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	55%	Pass	Stable memory usage

Step 4: Observations & Analysis

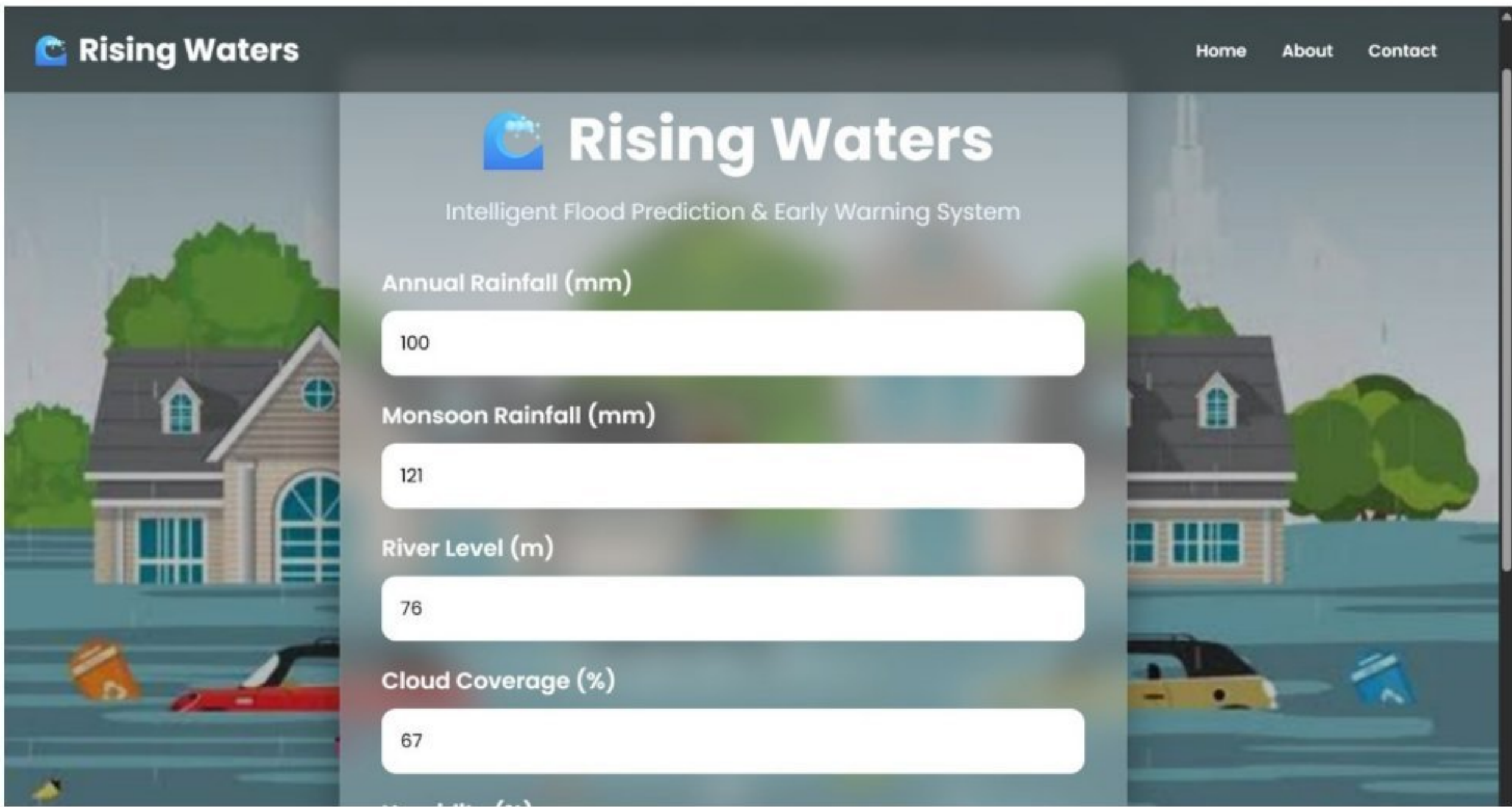
- The application successfully handled multiple prediction requests without failures.
- Average response time remained below **2 seconds**, meeting the performance objective.
- Input validation prevented invalid data from affecting the prediction process.
- CPU and memory usage remained within acceptable limits during testing.
- Overall, the application performed reliably and is suitable for small to medium-scale deployment.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Step 5: Screenshots / Evidence

Screenshot 1 - Rising Waters Home Page



Screenshot 2 - Current Weather Parameters

